

# Velora - Architecture and Delivery Plan

Version: 1.0 | Established: 2026-10-01 (Asia/Kolkata)

Requirements authority: VELORA\_MASTER.md. Progress and Duck protocol: SESSION\_LOG.md.

Intended repository location: docs/ARCHITECTURE.md.

Status: proposed engineering baseline; no source repository inspected or implementation verified.

## 1. Architecture approach

Build one modular FastAPI backend, one Next.js frontend and PostgreSQL. Begin with the manual approval workflow; make AI an optional draft-prefill service. Avoid microservices and framework complexity without a demonstrated need. Code 2 gets no active routes, screens, workers or tables yet; reuse stable episode and identity boundaries later.

| Component     | Baseline                                       | Responsibility                                                          |
|---------------|------------------------------------------------|-------------------------------------------------------------------------|
| Web           | Next.js, TypeScript, Tailwind CSS              | Role-specific mobile-friendly interfaces                                |
| API           | Python, FastAPI                                | Authentication integration, access policy, business rules, transactions |
| Persistence   | PostgreSQL; proposed SQLAlchemy + Alembic      | Relational data, constraints and versioned migrations                   |
| Files         | Private encrypted object storage               | Originals and authorised exports                                        |
| Extraction    | PyMuPDF + selected OCR + structured-output LLM | Source-linked draft extraction; no publication power                    |
| Jobs          | Redis + Celery when background work begins     | Extraction, reminder dispatch, retries                                  |
| Notifications | In-app initially                               | Delivery records; later external channels by explicit milestone         |

Exact package versions must be selected and pinned at setup after checking current compatibility. Authentication provider, OCR/model, storage and deployment provider remain open. Do not force a vendor just because it appears in a sample.

## 2. Proposed repository structure

```
velora/  
  frontend/  
    src/app/  
    src/features/  
    src/lib/  
  backend/  
    app/  
      api/  
      core/  
      models/  
      schemas/  
      policies/  
      services/  
      workers/  
    migrations/  
    tests/  
  docs/  
    VELORA_MASTER.md  
    ARCHITECTURE.md  
    SESSION_LOG.md
```

Routes handle validated requests/responses. Policies enforce scope and permissions. Services own workflows and transactions. Models persist facts. Schemas define external contracts. Workers invoke the same services and recheck current authorisation/version, rather than bypassing controls.

### 3. Domain model

| Entity                          | Purpose and essential relationships                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------|
| User                            | Identity; do not encode all access as one global role                                                |
| Clinic                          | Tenant and operating settings                                                                        |
| ClinicMembership                | User + clinic + role(s) + active/verified status                                                     |
| Invitation                      | Intended identity, purpose, clinic/episode scope, expiry, consumed/revoked status; store token hash  |
| PatientProfile                  | Clinic-scoped patient record; optional link to verified user; no automatic merge by name/phone alone |
| RecoveryEpisode                 | Patient + clinic + discharge date + status + assigned team                                           |
| CareTeamAssignment              | Episode + staff membership + responsibility and active interval                                      |
| CaregiverAccess                 | Patient/episode + caregiver + explicit scopes + grant/revocation history                             |
| Document                        | Episode + private object key + type + checksum + validation/extraction status                        |
| CarePlanVersion                 | Episode + version + draft/approved/superseded state + approver/time + source associations            |
| CarePlanItem                    | Structured clinical instruction + original text + source location + clarification/review status      |
| MedicationSchedule              | Approved version/item + exact clinician-confirmed schedule/conditions                                |
| Task / TaskLog                  | Scheduled item and appendable actor-attributed self-report                                           |
| CheckInTemplate / Response      | Clinic-approved versioned questions and dated responses                                              |
| ClarificationRequest / Response | Episode + owner + status + clinical/operational routing + history                                    |
| Notification / DeliveryAttempt  | Recipient, deduplication key, version, attempts, delivery and acknowledgement facts                  |
| AuditEvent                      | Actor, tenant, object, action, outcome, timestamp; avoid raw clinical text                           |

Create entities only as their milestones require them. Avoid duplicated data models for Code 2 in advance. Use foreign keys/uniqueness and scoped queries to prevent cross-tenant associations. Keep timestamp instants in UTC with an explicit scheduling timezone; render in the user's context.

### 4. State machines and transaction boundaries

**Episode:** draft > active after approved plan > completed > archived. Closure is operational; unresolved clinical issues require review. Reopening, if needed, must be an explicit audited operation with reminder reconciliation.

**Document:** uploaded > validating > ready for extraction > extracting > extracted / needs clarification / failed. Invalid or unsafe files go to rejected/quarantined state. A retry does not duplicate plans or files. A partially successful AI extraction cannot activate instructions.

**Plan:** draft > in review > approved; review may return to needs clarification/draft. Update creates a NEW draft while old approved version remains active until replacement approval. In one transaction approve the new version, supersede the previous current version, record audit, and enqueue downstream work reliably. Prevent concurrent double approval with optimistic locking or row locking and a current-version constraint.

**Request:** open > assigned > answered > closed; reopen is explicit and audited. Delivery is not acknowledgement; acknowledgement is not resolution.

**Logs:** append corrections with attribution; do not silently alter reported medication history. Distinguish reported occurrence time from recorded-at time.

## 5. Access policy

Policy decision inputs: authenticated identity, active clinic membership or patient ownership, assignment/caregiver grant, requested operation, target object and state. Resolve clinic scope server-side; never trust a client-supplied role or clinic ID alone.

- Deny access by default, including list/search/count/export endpoints.
- Admin may configure roles but cannot browse clinical content solely as admin or self-certify clinical eligibility.
- Caregiver revocation invalidates future requests and jobs. Existing downloaded copies cannot be recalled; short-lived links reduce remaining exposure.
- Signed downloads require fresh access checks; keep expiration short and objects private.
- Multi-role permissions are bounded by clinic and record assignment.
- Workers recheck approved-plan version and active access before delivery.
- Session/token strategy must include revocation, expiry, rate limits, secure cookie/CSRF protections where applicable, and staff MFA before real-data pilot.

## 6. Proposed API contracts

Paths are planning examples, not existing endpoints. For each implemented route document schema, actor, policy, validation errors and relevant VEL IDs.

| Endpoint                                  | Action / gate                                                  |
|-------------------------------------------|----------------------------------------------------------------|
| POST /clinics/{id}/staff-invitations      | Own-clinic admin; eligible role assignment process             |
| POST /invitations/accept                  | Bound verified identity; unexpired unused token                |
| POST /patients                            | Authorised clinic staff; identity duplication check            |
| POST /episodes                            | Authorised staff; patient in same clinic; team assignment      |
| GET /episodes/{id}                        | Own/shared/assigned record                                     |
| POST /episodes/{id}/documents             | Upload permission; validated scoped intake                     |
| POST /episodes/{id}/care-plans            | Authorised staff creates draft                                 |
| PATCH /care-plans/{id}/draft              | Editable version and expected revision required                |
| POST /care-plans/{id}/approve             | Assigned verified reviewer; unresolved clinical fields blocked |
| GET /episodes/{id}/care-plan              | Patient gets current approved content, not unpublished drafts  |
| POST /episodes/{id}/caregiver-invitations | Patient/verified representative grants selected scopes         |
| DELETE /caregiver-access/{id}             | Authorised revocation with audit                               |
| POST /tasks/{id}/logs                     | Authorised actor; state and duplicate check                    |

| Endpoint                       | Action / gate                                          |
|--------------------------------|--------------------------------------------------------|
| POST /episodes/{id}/check-ins  | Authorised response to current template                |
| POST /episodes/{id}/questions  | Own/shared/assigned; records requester                 |
| POST /questions/{id}/responses | Assigned staff; clinical replies require reviewer role |
| POST /episodes/{id}/close      | Staff policy; pending-item review and recorded reason  |

Use structured error codes for forbidden access, stale revision, incomplete review, invalid upload and processing failure. Do not leak record existence through error text. Mutation retry behaviour must be explicit: approval, invitations and job submission need idempotency/concurrency protection.

## 7. AI and document pipeline

Validate upload > private storage > text/OCR extraction with page positions > bounded schema extraction > deterministic validation > source-linked draft > clinical review. File contents are data, never executable instructions or tool-authorisation prompts. The LLM gets no approval, prescription, unrestricted network or production mutation authority.

Preserve original medication strings, units and conditions. Flag contradictions and missing fields. Do not normalise ambiguous frequencies/durations into invented values. Translation/simplification stays attached to the plan version and requires review. Keep document/model access scoped; evaluate vendor retention before real data. Manual draft creation remains available on AI failure. No real clinical payloads in routine logs or fixtures.

## 8. Reminders and reliability

Schedule only approved, explicit instructions. Deduplicate per schedule occurrence, recipient and channel. Before sending, confirm episode is active, version current, instruction still applicable and recipient authorised. Cancel stale pending work when a plan changes or episode closes. Use retry limits and failure visibility; a failed job cannot become a silent success. Respect medication timing rather than applying blanket quiet-hours shifts to prescribed schedules. No response is not recorded, not inferred nonadherence.

## 9. Delivery milestones and exit checks

| Milestone | Deliverable                                               | Exit evidence                                                                                                          |
|-----------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| M0        | These documents, decisions backlog, synthetic sample plan | Scope and traceability ready; open decisions explicit                                                                  |
| M1        | Manual vertical slice                                     | Real end-to-end synthetic journey, isolated records, draft/approval gate verified                                      |
| M2        | AI-assisted draft extraction                              | Field/source checks on representative synthetic/de-identified authorised samples; ambiguity and failure cases reviewed |
| M3        | Caregiver, medication/tasks and reminders                 | Grant/revoke, attribution, schedule update and duplicate prevention checks                                             |
| M4        | Check-ins, questions, dashboard, basic follow-up          | Owner assignment, clinical routing, reopening and response-hours behaviour verified                                    |
| M5        | Code 1 pilot readiness                                    | Staff MFA, audited boundaries, restore test, privacy/provider review, clinical workflow review, deployment decision    |
| Deferred  | Code 2 integration                                        | Only after explicit user go-ahead; update requirements first                                                           |

M0 does not imply clinic validation or repo setup is complete. No calendar promise has been made. Avoid feature implementation out of order unless needed for a dependency or explicitly requested.

## 10. Meaningful verification

Check cross-clinic reads/writes/files, unassigned staff, caregiver revocation, coordinator approval denial, admin clinical-access denial, drafts hidden from patients, stale/concurrent approvals, source ambiguity, reminder supersession/duplication, conditional medication behaviour, document rejection and episode closure. Use unit tests for policy/state logic, database integration tests for isolation/transactions, and one end-to-end synthetic user journey per milestone. Evaluate AI extraction separately from software passing tests. Record exact commands and observed outcomes; never manufacture evidence.

## 11. Implementation method and change decisions

For every feature derive: user action > entity > permission > API > transaction/service > UI > failure path > acceptance evidence. Explain code structure, responsibilities and important lines to Swayam. Keep changes focused and reversible, preserve user work, and inspect repository instructions before editing.

Proposed decisions can be refined through ordinary engineering judgment if requirements remain intact. Record significant changes with ID/date/problem/options/decision/consequences. Scope changes, clinical-policy choices and destructive migrations need explicit resolution; do not silently label them implementation details.

### Initial decisions

- ADR-001: modular monolith; proposed baseline, no implementation verified.
- ADR-002: relational database for tenant/episode/version relationships; proposed baseline.
- ADR-003: manual approval flow before AI prefilling; agreed delivery sequence.
- ADR-004: private source documents and source-linked draft fields; baseline requirement.
- ADR-005: exact auth/model/OCR/storage vendors remain open until setup and data-handling review.